Diploma in Engineering and Technology • Semester 1 • Basic Science

Applied Chemistry-I Downloadable Notes

Student-friendly digital textbook + workbook + revision guide + answer key. Content is built from the official 1004 syllabus: atomic structure, chemical bonding, analytical chemistry, water treatment, engineering materials, electrochemistry and corrosion.

1004 Subject code	**3** Credits	**45** Periods

Mini Formula Bank

Chemistry

$\lambda = h/mv$ de Broglie	$pH = -\log[H^+]$ Acidity
$N_1V_1=N_2V_2$ Titration	$m = ZIt$ Electrolysis

How to write answers

Exam friendly

1. Write definition in the first line.
2. Add principle / equation / labelled diagram where needed.
3. Explain one real engineering application.
4. For numericals: Given → Formula → Substitution → Answer.

Course Overview

Atomic structure → Chemical bonding → Solutions → pH & Titration → Water treatment → Engineering materials → Polymers → Nanomaterials → Electrochemistry → Corrosion

Course objectives

- Understand atomic structure, chemical bonding, water treatment, engineering materials, electrochemistry and corrosion.
- Apply chemistry principles to domestic and industrial engineering problems.
- Select suitable materials and treatment methods for practical engineering use.

Course outcomes

CO	Outcome	Hours	Level
CO1	Explain atomic structure and chemical bonding.	8	Understandi
CO2	Apply analytical chemistry fundamentals to solve engineering problems and understand water treatment methods.	14	Applying
CO3	Explain engineering materials for domestic and industrial applications.	9	Understandi
CO4	Apply electrochemistry and corrosion concepts to solve engineering problems.	12	Applying

Module map

Daily study method

1. Read one lesson slowly.
2. Copy formulas with symbol meaning and units.
3. Redraw the chemistry sketch.
4. Solve one numerical or example.
5. Check answer key after trying.

Numerical format

1. Given data with units.
2. Formula used.
3. Unit conversion.
4. Substitution and calculation.
5. Final answer with unit.

Long answer format

1. Definition.
2. Principle/working.
3. Labelled diagram or table.
4. Example.
5. Application / advantage / limitation.

Mini Formula Bank

Only chemistry formulas from Applied Chemistry-I are included here. Use this for numericals, titration, pH and electrochemistry problems.

Search formula, topic or use...

de Broglie wavelength

$\lambda = h / mv$

Symbols: λ wavelength, h Planck constant, m mass, v velocity

Use: Used to calculate matter wave wavelength of a moving particle.

> **Careful:** Use SI units: kg and m/s.

Heisenberg uncertainty

$\Delta x \cdot \Delta p \geq h / 4\pi$

Symbols: Δx uncertainty in position, Δp uncertainty in momentum

Use: Shows why exact position and momentum cannot both be measured exactly.

> **Careful:** Momentum uncertainty may be mΔv.

Molarity

M = number of moles / volume in litre

Symbols: mol L^{-1}

Use: Concentration of solution in volumetric analysis.

> **Careful:** Volume must be in litres.

Number of moles

n = given mass / molar mass

Symbols: n in mol

Use: Needed before molarity calculation.

> **Careful:** Use correct molar mass from formula.

Normality

N = gram equivalents / volume in litre

Symbols: eq L^{-1}

Use: Useful in acid-base titration calculations.

> **Careful:** Equivalent mass = molar mass / valency factor.

ppm

ppm = mg of solute / litre of solution

Symbols: For dilute aqueous solutions

Use: Used for hardness, impurities and water quality.

> **Careful:** 1 ppm ≈ 1 mg/L in water.

pH

$pH = -\log_{10}[H^+]$

Symbols: $[H^+]$ in mol L^{-1}

Use: Measure of acidity.

> **Careful:** Low pH means high acidity.

pOH

$pOH = -\log_{10}[OH^-]$

Symbols: $[OH^-]$ in mol L^{-1}

Use: Measure of basicity.

> **Careful:** At 25°C, pH + pOH = 14.

Normality equation

$N_1V_1 = N_2V_2$

Symbols: For acid-base titration at equivalence point

Use: Find unknown concentration from titration data.

> **Careful:** Use same volume unit on both

Faraday's first law

$m = ZIt$

Symbols: m mass deposited, Z electrochemical equivalent, I current, t time

Use: Mass deposited during electrolysis.

> **Careful:** Convert time to seconds.

sides.

Faraday constant method

$m = (EIt) / F$

Symbols: E equivalent mass, F = 96500 C mol^{-1} eq^{-1}

Use: Electrolysis numerical problems.

> **Careful:** Equivalent mass = atomic mass / valency.

Cell EMF

$E_{cell} = E_{cathode} - E_{anode}$

Symbols: Electrode potentials in volt

Use: Find emf of electrochemical cell.

> **Careful:** Cathode is reduction electrode.

Water hardness as $CaCO_3$

$Hardness = (\text{mass of } CaCO_3 \text{ equivalent} / \text{volume of water}) \times 10^6$

Symbols: ppm

Use: Express water hardness in common engineering unit.

> **Careful:** Convert all values consistently.

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Module 1 • CO1 • 8 hours

Atomic Structure and Chemical Bonding

This module builds the foundation for understanding how atoms are arranged, why electrons occupy specific orbitals, and how atoms combine through chemical bonds.

ഈ module atomic structure, orbital concept, electronic configuration, chemical bonding എന്നിവയ്ക്ക് base നൽകുന്നു. Chemistry-ൽ later topics മനസ്സിലാക്കാൻ ഇതാണ് foundation.

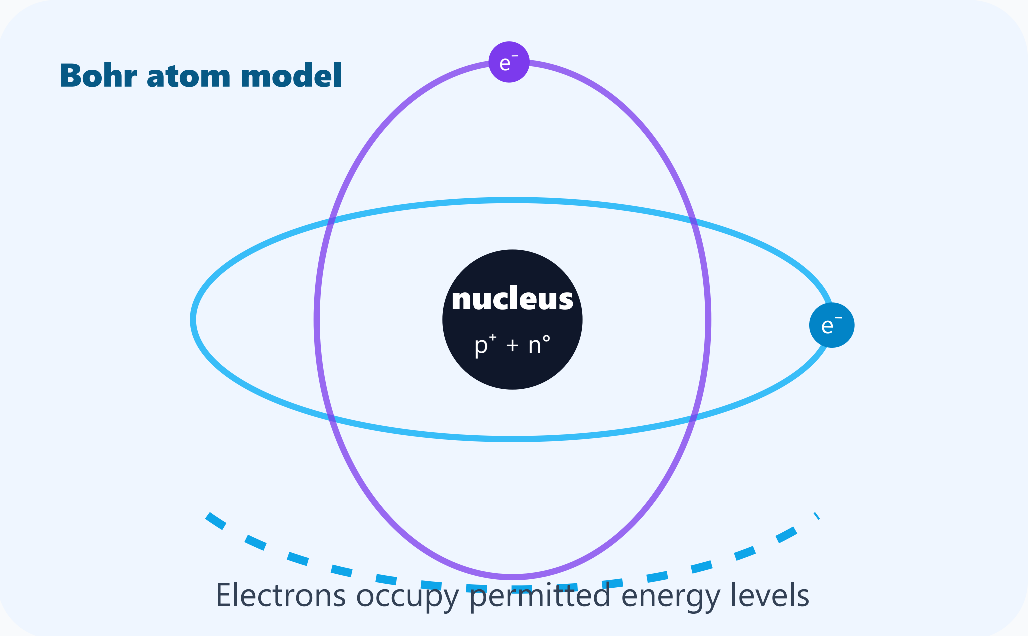
M1.01 Structure of atom and Bohr's atom model

English explanation

An atom is the smallest unit of an element that shows the chemical properties of that element. It contains a positively charged nucleus with protons and neutrons, and negatively charged electrons arranged around the nucleus. In Bohr's model, electrons revolve only in certain permitted circular orbits called shells or energy levels. As long as an electron remains in a permitted orbit, it does not radiate energy. Energy is absorbed when an electron jumps to a higher orbit and emitted when it returns to a lower orbit. This model explains stability of atoms and line spectrum of hydrogen, but it cannot explain spectra of multi-electron atoms or fine structure of spectral lines.

Malayalam support

Atom എന്നത് ഒരു element-ന്റെ chemical property കാണിക്കുന്ന ഏറ്റവും ചെറിയ unit ആണ്. Nucleus-ൽ proton, neutron എന്നിവ ഉണ്ടാകും; electron-കൾ nucleus-നെ ചുറ്റി shells/energy levels-ൽ കാണപ്പെടുന്നു. Bohr model പ്രകാരം electron-കൾ allowed orbit-ൽ ഇരിക്കുമ്പോൾ energy radiate ചെയ്യില്ല. Electron higher orbit-ലേക്ക് പോകുമ്പോൾ energy absorb ചെയ്യുകയും lower orbit-ലേക്ക് മടങ്ങുമ്പോൾ energy emit ചെയ്യുകയും ചെയ്യും. Hydrogen spectrum explain ചെയ്യാൻ ഇത് സഹായിച്ചെങ്കിലും multi-electron atoms explain ചെയ്യാൻ limitation ഉണ്ട്.



Important points

- Nucleus contains protons and neutrons.
- Electrons occupy permitted energy levels called shells.
- Bohr model explains atomic stability and hydrogen spectrum.
- Main limitation: not suitable for complex atoms.

Examples

- Hydrogen atom has one proton and one electron.
- Sodium has electronic configuration 2,8,1 and easily forms Na^+.

Applications

- Understanding valency and bonding.
- Explaining emission spectra used in analysis and lamps.

Study tricks / exam tips:

- For exam: write postulates first, then merits and demerits separately.
- Do not write Bohr radius/energy derivation because syllabus omits expression.

M1.02 Dual nature, de Broglie equation and uncertainty principle

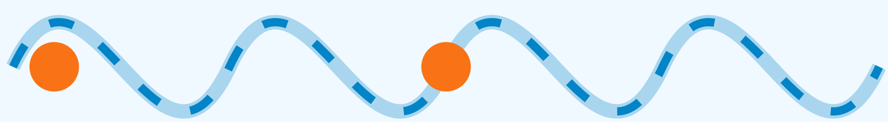
English explanation

Light and matter show dual nature: they can behave as waves and particles depending on the experiment. de Broglie proposed that every moving particle has an associated matter wave. Its wavelength is inversely proportional to momentum, so very small particles like electrons show wave nature clearly, while ordinary objects have extremely small wavelengths. Heisenberg’s uncertainty principle says that it is impossible to determine both the exact position and exact momentum of a microscopic particle at the same time. This is not due to instrument defect; it is a fundamental limitation in the microscopic world.

Malayalam support

Light-നും matter-നും dual nature ഉണ്ട്: ചില സാഹചര്യങ്ങളിൽ wave പോലെ, ചിലപ്പോൾ particle പോലെ behave ചെയ്യും. de Broglie പ്രകാരം moving particle-ന് matter wave ഉണ്ടാകും. Momentum കൂടുതലായാൽ wavelength കുറയും. Electron പോലുള്ള small particles-ൽ wave nature വ്യക്തമാണ്. Heisenberg uncertainty principle പ്രകാരം ഒരു microscopic particle-ന്റെ exact position ഉം exact momentum ഉം ഒരേ സമയം കൃത്യമായി അറിയാൻ കഴിയില്ല. ഇത് instrument error അല്ല; microscopic level-ലെ basic limitation ആണ്.

de Broglie matter wave



Higher momentum → smaller wavelength

$$\lambda = h / mv$$

Important points

- de Broglie equation: $\lambda = h/mv$.
- Heisenberg principle: $\Delta x \, \Delta p \geq h/4\pi$.
- Matter waves are significant for electrons and other microscopic particles.
- Uncertainty principle is important in quantum chemistry.

Examples

- Electron diffraction proves wave nature of electrons.
- A cricket ball has matter wave, but its wavelength is too small to observe.

Applications

- Electron microscope.
- Understanding orbitals and quantum model of atom.

Study tricks / exam tips:

- Numerical steps: write formula, convert mass and velocity to SI, substitute, final wavelength in metre.
- Do not confuse de Broglie wavelength with ordinary light wavelength.

M1.03 Orbitals, quantum numbers and electronic configuration

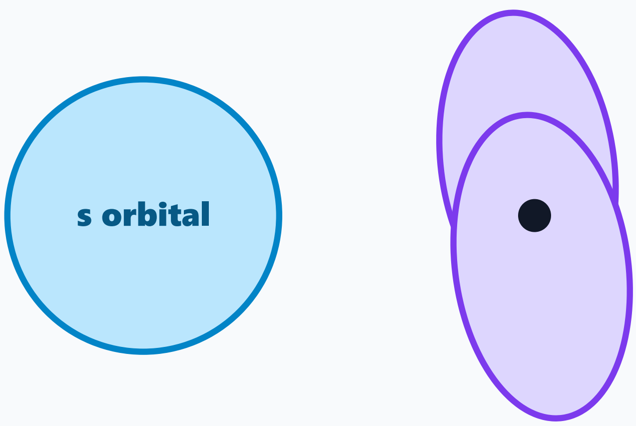
English explanation

In modern atomic theory, an orbital is a three-dimensional region around the nucleus where the probability of finding an electron is maximum. Orbitals are described using quantum numbers. The principal quantum number n gives shell, azimuthal quantum number l gives subshell, magnetic quantum number m gives orientation, and spin quantum number s gives spin of electron. Electronic configuration is written using Aufbau principle, Pauli exclusion principle and Hund’s rule. Aufbau principle says electrons fill lower energy orbitals first. Pauli principle says an orbital can contain maximum two electrons with opposite spins. Hund’s rule says electrons occupy degenerate orbitals singly before pairing.

Malayalam support

Modern atomic theory-ൽ orbital എന്നത് electron കണ്ടെത്താനുള്ള probability ഏറ്റവും കൂടുതലുള്ള 3D region ആണ്. Quantum numbers orbital-ന്റെ details പറയുന്നു: n shell, l subshell, m orientation, s spin. Electronic configuration എഴുതുമ്പോൾ Aufbau principle, Pauli exclusion principle, Hund’s rule എന്നിവ follow ചെയ്യണം. Low energy orbital ആദ്യം fill ചെയ്യണം. ഒരു orbital-ൽ maximum രണ്ട് electron-കൾ opposite spin-ൽ മാത്രം. Equal energy orbitals-ൽ ആദ്യം singly fill ചെയ്ത് ശേഷം pairing നടക്കും.

Orbitals and probability regions



maximum probability region   p orbital has two lobes

Important points

- s orbital holds 2 electrons; p holds 6; d holds 10; f holds 14.
- Aufbau order begins 1s, 2s, 2p, 3s, 3p, 4s, 3d.
- Pauli: no two electrons in an atom have the same four quantum numbers.
- Hund: maximum multiplicity before pairing.

Examples

- Carbon: $1s^2\ 2s^2\ 2p^2$.
- Calcium: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$.

Applications

- Predicting valency and bonding behaviour.
- Understanding periodic properties.

Study tricks / exam tips:

- Learn first 20 elements by shell configuration: 2, 8, 8, 2 pattern helps.
- In exam, write both orbital notation and shell distribution if asked.

M1.04

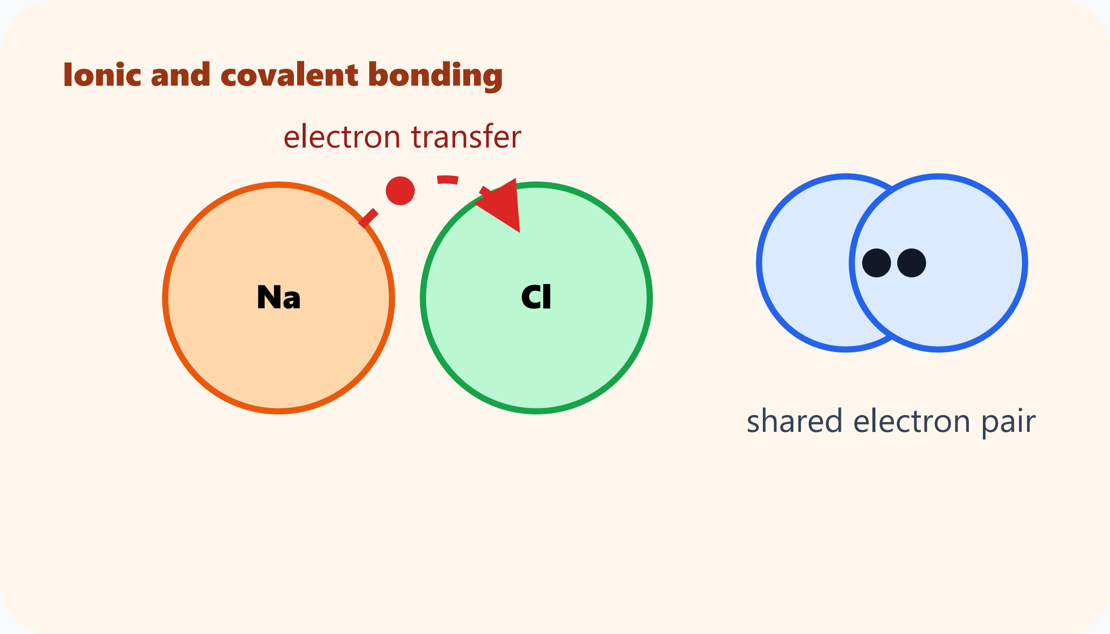
Chemical bonding: ionic, covalent, coordinate and hydrogen bonding

English explanation

Chemical bonding is the attractive force that holds atoms together in molecules or crystals. Atoms combine to attain stable electronic configuration, often octet configuration. Ionic bond is formed by transfer of electrons, as in NaCl. Covalent bond is formed by sharing of electrons, as in H_2 and HF. Coordinate bond is a covalent bond in which the shared pair is donated by one atom, as in NH_4^+. Hydrogen bonding is a weak attraction involving hydrogen bonded to highly electronegative atoms like F, O or N. Hydrogen bonding explains anomalous behaviour of water such as high boiling point, high surface tension and lower density of ice.

Malayalam support

Chemical bond എന്നത് atoms-നെ molecule/crystal-ൽ പിടിച്ചു നിർത്തുന്ന attractive force ആണ്. Stable electronic configuration നേടാൻ atoms combine ചെയ്യുന്നു. Electron transfer വഴി ionic bond ഉണ്ടാകും; ഉദാ: NaCl. Electron sharing വഴി covalent bond ഉണ്ടാകും; ഉദാ: H_2, HF. Shared pair ഒരു atom donate ചെയ്യുമ്പോൾ coordinate bond; ഉദാ: NH_4^+. H atom F/O/N പോലുള്ള electronegative atom-നോട് bonded ആയാൽ hydrogen bonding ഉണ്ടാകും. Water-ന്റെ high boiling point, ice density കുറവ് തുടങ്ങിയ anomalous behaviour ഇതിലൂടെ explain ചെയ്യാം.



Important points

- Octet rule: atoms tend to attain eight electrons in valence shell.
- Ionic bond: electron transfer and electrostatic attraction.
- Covalent bond: sharing of electron pair.
- Coordinate bond: donor atom supplies both shared electrons.
- Hydrogen bonding: intermolecular or intramolecular attraction.

Examples

- $Na + Cl \rightarrow Na^{+}Cl^{-}$.
- H_2 has one shared pair.
- NH_3 donates lone pair to H^{+} to form NH_4^{+}.

Applications

- Explaining properties of salts, polymers and water.
- Choosing materials based on bonding and structure.

Study tricks / exam tips:

- Bond-type answers should include definition + formation + example + property.
- For H_2O, mention hydrogen bonding and tetrahedral bent arrangement only briefly.

Solved Examples

Worked examples are included in the module itself. Master Answer Key contains answers for expected questions.

de Broglie wavelength of electron

Given: Mass of electron = 9.1×10^{-31} kg, velocity = 2.0×10^{6} m/s, h = 6.626×10^{-34} J s.

$\lambda = h/mv$

$\lambda = 6.626\times10^{-34} / (9.1\times10^{-31} \times 2.0\times10^{6})$

$\lambda = 3.64 \times 10^{-10}$ m

de Broglie wavelength = 3.64×10^{-10} m.

Common mistake: Keep mass in kg and velocity in m/s.

Electronic configuration of calcium

Given: Atomic number of Ca = 20.

Fill orbitals in Aufbau order.

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$

Shell distribution = 2, 8, 8, 2

Ca: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$.

Common mistake: 4s fills before 3d for first 20 elements.

Identify bond type

Given: NaCl, H_2, NH_4^+ and H_2O.

NaCl: electron transfer.

H_2: electron sharing.

NH_4^+: lone pair donation by NH_3 to H^+.

H_2O: hydrogen bonding between molecules.

NaCl ionic, H_2 covalent, NH_4^+ coordinate, H_2O hydrogen bonding.

Common mistake: Use formation method to identify bond type.

Expected Questions

Questions only are shown here. Full answers are available in Master Answer Key.

Q1 Explain Bohr's atom model. Write its important postulates, merits and demerits.

Q2 Define de Broglie wavelength and solve a simple problem using $\lambda = h/mv$.

Q3 State Heisenberg's uncertainty principle and explain its meaning.

Q4 Explain quantum numbers and their significance.

Q5 Write Aufbau principle, Pauli exclusion principle and Hund's rule.

Q6 Write electronic configuration of first 20 elements using examples.

Q7 Compare ionic, covalent and coordinate bonds with examples.

Q8 Explain anomalous behaviour of water due to hydrogen bonding.

Electronic Configuration of First 20 Elements

Useful for M1.03 electronic configuration practice.

Element	Atomic No.	Orbital configuration	Shell distribution
H	1	$1s^1$	1
He	2	$1s^2$	2
Li	3	$1s^2\ 2s^1$	2,1
Be	4	$1s^2\ 2s^2$	2,2
B	5	$1s^2\ 2s^2\ 2p^1$	2,3
C	6	$1s^2\ 2s^2\ 2p^2$	2,4
N	7	$1s^2\ 2s^2\ 2p^3$	2,5
O	8	$1s^2\ 2s^2\ 2p^4$	2,6
F	9	$1s^2\ 2s^2\ 2p^5$	2,7
Ne	10	$1s^2\ 2s^2\ 2p^6$	2,8
Na	11	$1s^2\ 2s^2\ 2p^6\ 3s^1$	2,8,1

Element	Atomic No.	Orbital configuration	Shell distribution
Mg	12	$1s^2\ 2s^2\ 2p^6\ 3s^2$	2,8,2
Al	13	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$	2,8,3
Si	14	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$	2,8,4
P	15	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$	2,8,5
S	16	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$	2,8,6
Cl	17	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$	2,8,7
Ar	18	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$	2,8,8
K	19	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$	2,8,8,1
Ca	20	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$	2,8,8,2

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Module 2 • CO2 • 14 hours

Analytical Chemistry and Water Treatment

This module connects chemistry calculations with real engineering needs such as solution preparation, titration, pH control, hardness removal and potable water treatment.

ഈ module chemistry calculations engineering use-നോട് connect ചെയ്യുന്നു: solution preparation, titration, pH control, hardness removal, potable water treatment എന്നിവ പ്രധാനമാണ്.

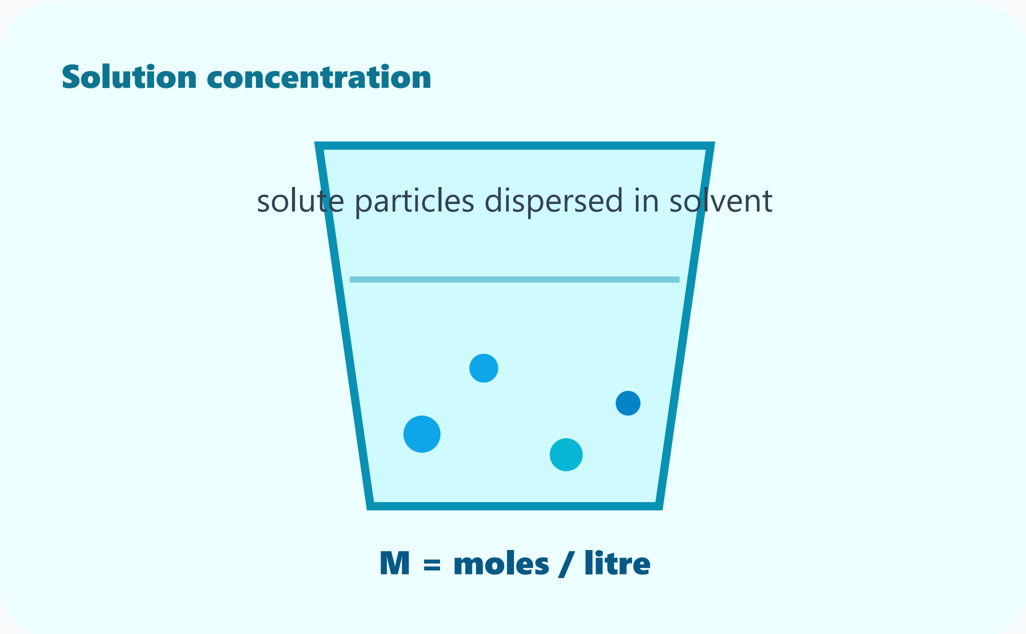
M2.01 Solutions and concentration: molarity, normality and ppm

English explanation

A solution is a homogeneous mixture of solute and solvent. The solute is the substance dissolved and the solvent is the medium that dissolves it. In chemistry and engineering, concentration tells how much solute is present in a given quantity of solution. Molarity is moles of solute per litre of solution. Normality is gram equivalents of solute per litre of solution. ppm means parts per million and is commonly used for very dilute solutions such as water impurities and hardness. Concentration calculations are important in titration, water treatment, electroplating bath preparation and industrial chemical dosing.

Malayalam support

Solution എന്നത് solute solvent-ൽ dissolve ചെയ്ത് ഉണ്ടാകുന്ന homogeneous mixture ആണ്. Solute dissolve ചെയ്യുന്ന substance, solvent dissolve ചെയ്യിക്കുന്ന medium. Concentration solution-ൽ solute എത്രയുണ്ട് എന്ന് പറയുന്നു. Molarity = litre-ൽ moles. Normality = litre-ൽ gram equivalents. ppm dilute solutions-ൽ, പ്രത്യേകിച്ച് water quality-ൽ ഉപയോഗിക്കുന്നു. Titration, water treatment, electroplating bath, chemical dosing എന്നിവയ്ക്ക് concentration calculation വളരെ പ്രധാനമാണ്.



Important points

Examples

- M = moles / volume in litre.
- Normality depends on equivalent mass.
- ppm is useful for water quality and hardness.
- Always convert mL to L before calculation.

- 0.1 M NaOH means 0.1 mole NaOH in 1 litre solution.
- 50 ppm chloride means about 50 mg chloride per litre of water.

Applications

- Preparation of standard solution.
- Chemical dosing in boilers and treatment plants.

Study tricks / exam tips:

- Molarity problem format: mass → moles → volume in litre → M.
- Normality problem format: mass → equivalents → volume → N.

M2.02 pH, pOH, buffer solutions and applications of pH

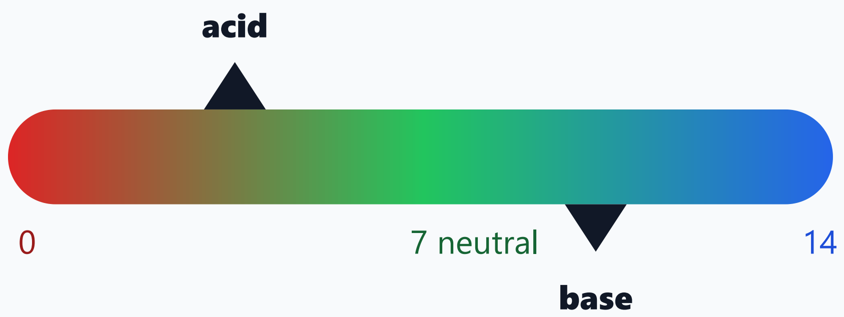
English explanation

pH is a logarithmic measure of hydrogen ion concentration. A solution with pH less than 7 is acidic, pH 7 is neutral and pH greater than 7 is basic at 25°C. pOH measures hydroxide ion concentration and pH + pOH = 14 for aqueous solutions at 25°C. Buffer solutions resist change in pH when small amounts of acid or base are added. Acidic buffer contains a weak acid and its salt, for example acetic acid and sodium acetate. Basic buffer contains a weak base and its salt, for example ammonium hydroxide and ammonium chloride. pH control is important in water treatment, agriculture, medicine, electroplating and corrosion prevention.

Malayalam support

pH എന്നത് hydrogen ion concentration-ന്റെ logarithmic measure ആണ്. pH 7-ൽ താഴെ acidic, 7 neutral, 7-ൽ മുകളിൽ basic. pOH hydroxide ion concentration കാണിക്കുന്നു; 25°C-ൽ pH + pOH = 14. Buffer solution ചെറിയ acid/base ചേർത്താലും pH വലിയ മാറ്റം വരാതെ maintain ചെയ്യും. Acetic acid + sodium acetate acidic buffer; ammonium hydroxide + ammonium chloride basic buffer. Water treatment, agriculture, medicine, electroplating, corrosion control എന്നിവയിൽ pH control പ്രധാനമാണ്.

pH scale



$$pH = -\log_{10}[H^+]$$

Important points

- $pH = -\log[H^+]$.
- $pOH = -\log[OH^-]$.
- $pH + pOH = 14$ at 25°C.
- Buffers resist pH change.

Examples

- $[H^+] = 10^{-3}$ M gives pH = 3.
- Blood pH is maintained nearly constant by buffer systems.

Applications

- Controlling boiler water pH.
- Maintaining electroplating bath quality.

Study tricks / exam tips:

- Remember: acidic means lower pH, not higher pH.
- In log problems, 10^{-5} gives pH 5 directly.

M2.03 Volumetric analysis and titration

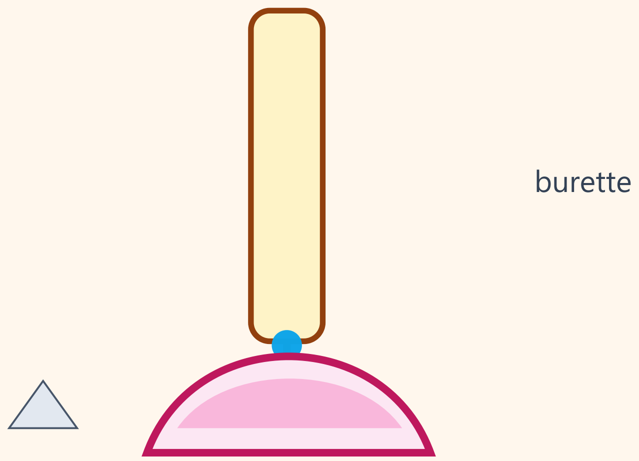
English explanation

Volumetric analysis is a quantitative method used to find the concentration of a solution by measuring the volume of another solution of known concentration required for complete reaction. In titration, the known solution is usually taken in a burette and the unknown solution in a conical flask. The end point is the stage at which the indicator shows completion of reaction by a colour change. The choice of indicator depends on the type of titration. For strong acid–strong base titration, phenolphthalein or methyl orange may be used. The normality equation $N_1V_1 = N_2V_2$ is used when acid and base react completely in equivalent amounts.

Malayalam support

Volumetric analysis concentration അറിയാത്ത solution-ന്റെ concentration കണ്ടെത്താൻ ഉപയോഗിക്കുന്ന quantitative method ആണ്. Known concentration solution burette-ൽ, unknown solution conical flask-ൽ എടുക്കുന്നു. Indicator colour change കാണിക്കുന്ന point ആണ് end point. Indicator തിരഞ്ഞെടുക്കുന്നത് titration type-നെ ആശ്രയിക്കും. Strong acid–strong base titration-ൽ phenolphthalein അല്ലെങ്കിൽ methyl orange ഉപയോഗിക്കാം. Acid-base equivalents equal ആയപ്പോൾ $N_1V_1 = N_2V_2$ ഉപയോഗിക്കുന്നു.

Acid-base titration



Endpoint colour change appears near equivalence point

Important points

- Standard solution: accurately known concentration.
- Burette measures volume delivered.
- Pipette transfers fixed volume.
- Indicator detects end point.
- Normality equation: $N_1V_1 = N_2V_2$.

Examples

- HCl vs NaOH titration using phenolphthalein.
- Hardness estimation uses EDTA titration in laboratories.

Applications

- Quality control in chemical industries.
- Water analysis and acid/base concentration measurement.

Study tricks / exam tips:

- Read lower meniscus at eye level.
- Rinse burette with titrant and pipette with solution to be pipetted.

M2.04

Hard water, water softening and municipal water treatment

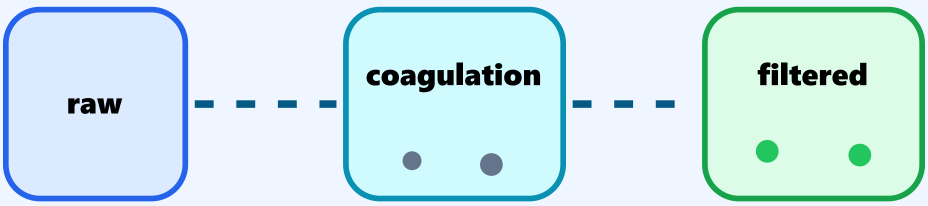
English explanation

Hard water does not easily produce lather with soap because calcium and magnesium ions react with soap to form insoluble scum. Temporary hardness is mainly due to bicarbonates of calcium and magnesium; permanent hardness is due to chlorides and sulphates. In boilers, hard water causes scale formation, reduced heat transfer, wastage of fuel and corrosion. Lime-soda process removes hardness by converting soluble calcium and magnesium salts into insoluble precipitates. Ion exchange method removes Ca^{2+} and Mg^{2+} ions using resins. Potable water must be colourless, odourless, free from harmful microorganisms and safe for drinking. Municipal treatment generally includes sedimentation, coagulation, filtration and sterilization by chlorination, bleaching powder or UV radiation.

Malayalam support

Hard water soap-നൊപ്പം lather ഉണ്ടാക്കാൻ ബുദ്ധിമുട്ടാണ്, കാരണം Ca^{2+}, Mg^{2+} ions soap-നോട് react ചെയ്ത് insoluble scum ഉണ്ടാക്കും. Temporary hardness calcium/magnesium bicarbonates കാരണം; permanent hardness chlorides/sulphates കാരണം. Boiler-ൽ scale, heat transfer കുറവ്, fuel waste, corrosion എന്നിവ ഉണ്ടാകും. Lime-soda process soluble salts-നെ insoluble precipitate ആക്കും. Ion exchange method resin ഉപയോഗിച്ച് Ca^{2+}, Mg^{2+} remove ചെയ്യും. Potable water drinking-ന് safe ആയിരിക്കണം. Municipal treatment-ൽ sedimentation, coagulation, filtration, sterilization എന്നിവ നടക്കുന്നു.

Water treatment sequence



softening / filtration / disinfection remove hardness and impurities

Important points

- Hardness causing ions: Ca^{2+} and Mg^{2+}.
- Boiler problems: scale and corrosion.
- Softening methods: lime-soda and ion exchange.
- Municipal water treatment: sedimentation → coagulation → filtration → sterilization.

Examples

- Kettles and boilers get scale deposits in hard water areas.
- Chlorination kills disease-causing microorganisms.

Applications

- Boiler feed water treatment.
- Drinking water purification and industrial process water.

Study tricks / exam tips:

- For water treatment answer, write each step in order and mention its purpose.
- Do not confuse filtration with sterilization: filtration removes suspended particles; sterilization kills microbes.

Solved Examples

Worked examples are included in the module itself. Master Answer Key contains answers for expected questions.

Molarity of NaOH solution

Given: 4 g NaOH dissolved in 500 mL solution. Molar mass NaOH = 40 g/mol.

Moles = 4/40 = 0.1 mol.

Volume = 500 mL = 0.5 L.

M = 0.1/0.5 = 0.2 M.

Molarity = 0.2 M.

Common mistake: Convert mL to L before using formula.

pH calculation

Given: $[H^+] = 1 \times 10^{-4}$ M.

$pH = -\log[H^+]$

$pH = -\log(10^{-4})$

pH = 4

pH = 4, so the solution is acidic.

Common mistake: 10^{-n} concentration gives pH n.

Titration using normality equation

Given: 25 mL of HCl is neutralized by 20 mL of 0.1 N NaOH.

$N_1V_1 = N_2V_2$

N(HCl) × 25 = 0.1 × 20

N(HCl) = 2/25 = 0.08 N

Normality of HCl = 0.08 N.

Common mistake: Use same volume unit on both sides.

Hardness reason

Given: Water sample gives poor lather with soap.

Presence of Ca^{2+}/Mg^{2+} ions is suspected.

These ions react with soap to form insoluble scum.

Softening removes these ions.

Poor lather is due to hardness caused by calcium and magnesium salts.

Common mistake: Mention scum formation in answer.

Expected Questions

Questions only are shown here. Full answers are available in Master Answer Key.

Q1 Define solute, solvent and solution. Explain molarity, normality and ppm.

Q2 Solve a molarity problem from mass and volume.

Q3 Explain pH, pOH and their relation. Add applications of pH.

Q4 What is buffer solution? Give one acidic and one basic buffer example.

Q5 Explain volumetric analysis, titration, end point and indicator.

Q6 Use normality equation to find unknown concentration.

Q7 Explain hardness of water and problems caused in boilers.

Q8 Explain lime-soda process, ion exchange method and municipal water treatment steps.

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Module 3 • CO3 • 9 hours

Engineering Materials, Polymers and Nanomaterials

This module explains materials used in engineering applications: alloys, glasses, refractories, polymers, rubbers and nanomaterials.

Engineering applications-ൽ ഉപയോഗിക്കുന്ന alloys, glasses, refractory materials, polymers, rubber, nanomaterials എന്നിവ ഈ module-ൽ പഠിക്കുന്നു.

M3.01 Alloys, glasses and refractory materials

English explanation

Engineering materials are selected according to strength, corrosion resistance, temperature resistance, cost and workability. An alloy is a metallic material formed by mixing two or more elements, at least one of which is a metal. Alloying improves properties such as hardness, strength, corrosion resistance and castability. Brass is an alloy of copper and zinc, used in fittings and decorative parts. Bronze is copper and tin, used in bearings and statues. Solder is mainly lead-tin or tin-based alloy used for joining metals. Glasses are amorphous inorganic materials. Sodium silicate glass is common glass, borosilicate glass resists thermal shock, safety glass reduces injury, and insulating glass reduces heat transfer. Refractory materials withstand high temperature without softening and are used in furnaces, kilns and boilers.

Malayalam support

Engineering materials തിരഞ്ഞെടുക്കുമ്പോൾ strength, corrosion resistance, temperature resistance, cost, workability എന്നിവ നോക്കണം. Alloy എന്നത് രണ്ട് അല്ലെങ്കിൽ കൂടുതൽ elements ചേർന്ന metallic material ആണ്. Alloying hardness, strength, corrosion resistance തുടങ്ങിയ properties improve ചെയ്യും. Brass = copper + zinc; Bronze = copper + tin; Solder metal joining-ന് ഉപയോഗിക്കുന്നു. Glass amorphous inorganic material ആണ്. Borosilicate glass thermal shock resist ചെയ്യും. Safety glass breakage injury കുറയ്ക്കും. Refractory materials high temperature-ൽ soften ചെയ്യാതെ നിലനിൽക്കും; furnaces, kilns, boilers എന്നിവയിൽ ഉപയോഗിക്കുന്നു.

Engineering materials

Cement | **Glass** | **Ceramic**

selection depends on property + application

Important points

- Purpose of alloying: improve strength, hardness, corrosion resistance, casting property.
- Brass: Cu + Zn; Bronze: Cu + Sn; Solder: joining alloy.
- Borosilicate glass resists thermal shock.
- Refractory materials must have high melting point and chemical stability.

Examples

- Brass taps and valves.
- Fire bricks inside furnaces.
- Borosilicate glass in laboratory beakers.

Applications

- Machine parts, electrical fittings, laboratory apparatus, furnace lining.

Study tricks / exam tips:

- Material answer method: composition → property → use.
- For refractory, always mention high-temperature application.

M3.02 Polymers, plastics and rubber

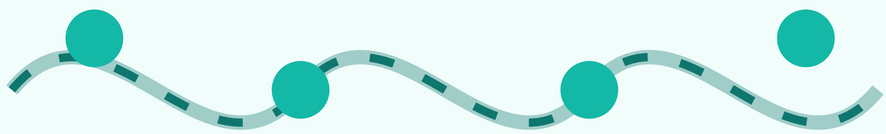
English explanation

A polymer is a large molecule formed by joining many small molecules called monomers. Polymerization is the process of forming polymer from monomers. Homopolymer is made from one type of monomer and copolymer from two or more monomers. Addition polymerization occurs without elimination of small molecules, as in polythene and PVC. Condensation polymerization occurs with elimination of small molecules such as water or HCl, as in Nylon-66 and Bakelite. Thermoplastics soften on heating and can be remoulded; thermosetting plastics become hard permanently and cannot be remoulded. Natural rubber is elastic but becomes sticky in summer and brittle in winter. Vulcanization with sulphur improves strength, elasticity and resistance. Buna-S and Buna-N are synthetic rubbers used in tyres, seals and oil-resistant products.

Malayalam support

Polymer എന്നത് monomer-കൾ ചേർന്ന് ഉണ്ടാകുന്ന large molecule ആണ്. Polymerization monomer-ൽ നിന്ന് polymer ഉണ്ടാക്കുന്ന process. Homopolymer ഒരു monomer type-ൽ നിന്ന്, copolymer രണ്ട് അല്ലെങ്കിൽ കൂടുതൽ monomer-ൽ നിന്ന്. Addition polymerization-ൽ small molecule eliminate ചെയ്യില്ല; ഉദാ: polythene, PVC. Condensation polymerization-ൽ water/HCl പോലുള്ള small molecule പുറത്തുപോകും; ഉദാ: Nylon-66, Bakelite. Thermoplastics heat ചെയ്യുമ്പോൾ soften ചെയ്ത് remould ചെയ്യാം. Thermosetting plastics permanent hard ആകും. Natural rubber sulphur ഉപയോഗിച്ച് vulcanize ചെയ്താൽ strength, elasticity, resistance improve ചെയ്യും. Buna-S, Buna-N synthetic rubbers ആണ്.

Polymer chain



repeating monomer units form long chains

Important points

- Monomer is the repeating unit.
- Addition polymers: polythene, PVC.
- Condensation polymers: Nylon-66, Bakelite.
- Thermoplastic: remouldable; thermosetting: not remouldable.
- Vulcanization improves rubber properties.

Examples

- Polythene bags and insulation.
- PVC pipes.
- Nylon fibres.
- Bakelite switches.

Applications

- Cable insulation, pipes, machine parts, tyres, seals and electrical fittings.

Study tricks / exam tips:

- Learn polymer examples with monomer + use as a pair.
- Do not call Bakelite a thermoplastic; it is thermosetting.

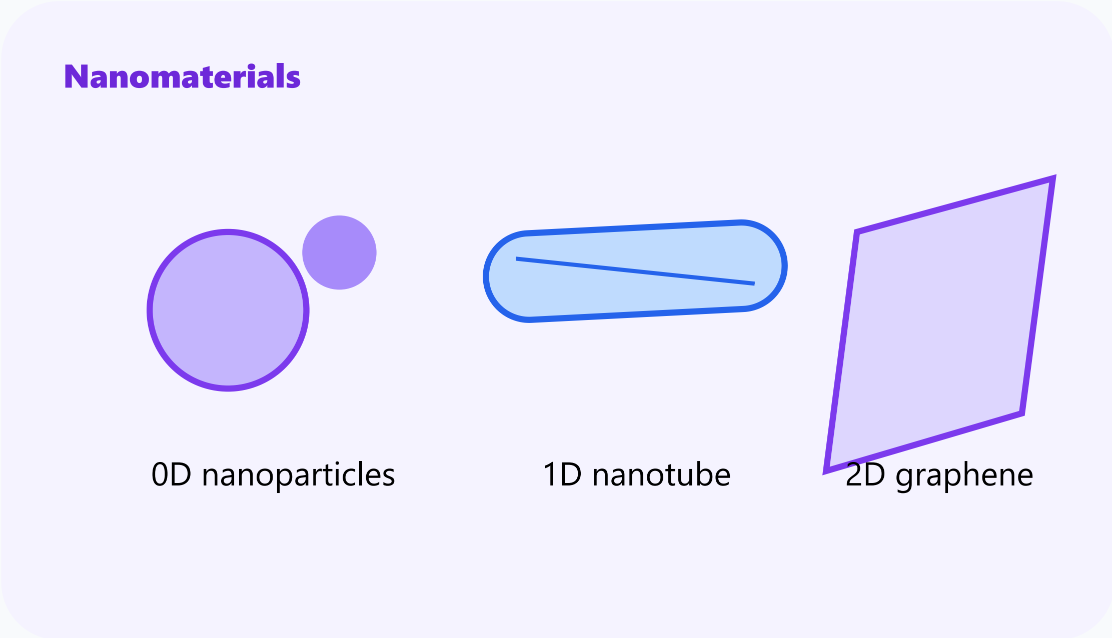
M3.03 Nanomaterials and nanotechnology

English explanation

Nanomaterials are materials with at least one dimension in the nanometre range, generally 1 to 100 nm. At nanoscale, materials may show properties different from bulk materials because of high surface area and quantum effects. Nanotechnology deals with design, production and application of materials at nanoscale. Based on dimension, nanomaterials can be classified as 0D, 1D and 2D. 0D materials have all dimensions in nanoscale, for example nanoparticles. 1D materials have two dimensions in nanoscale and one extended dimension, for example nanotubes and nanowires. 2D materials have one dimension in nanoscale, for example graphene sheets. Carbon nanotubes may be single-walled or multi-walled. Fullerenes are cage-like carbon molecules. Graphene is a single layer of carbon atoms arranged in hexagonal pattern.

Malayalam support

Nanomaterials-ൽ കുറഞ്ഞത് ഒരു dimension nanometre range-ൽ ആയിരിക്കും. Nanoscale-ൽ high surface area, quantum effects എന്നിവ കാരണം bulk material-നേക്കാൾ different properties കാണാം. Nanotechnology nanoscale materials design, production, application എന്നിവയെ കുറിച്ചാണ്. Dimension പ്രകാരം 0D, 1D, 2D ആയി classify ചെയ്യാം. 0D: nanoparticles; 1D: nanotubes/nanowires; 2D: graphene sheet. Carbon nanotubes SWCNT, MWCNT ആയി കാണാം. Fullerenes cage-like carbon molecules ആണ്. Graphene hexagonal pattern-ൽ carbon atoms ഉള്ള single layer ആണ്.



Important points

- Nanorange: about 1–100 nm.
- 0D example: nanoparticles.
- 1D example: carbon nanotubes.
- 2D example: graphene.
- Nanomaterials show high surface area and special properties.

Examples

- Silver nanoparticles show antimicrobial activity.
- CNTs have high strength and electrical conductivity.

Applications

- Sensors, coatings, electronics, medicine, batteries and composite materials.

Study tricks / exam tips:

- For classification, write dimension + example; do not write long unnecessary classification of CNT unless asked.
- Graphene is 2D, fullerene is 0D-like cage structure.

Solved Examples

Worked examples are included in the module itself. Master Answer Key contains answers for expected questions.

Choose material for laboratory beaker

Given: Beaker must resist sudden heating and cooling.

Thermal shock resistance is required.

Borosilicate glass has low expansion and good thermal resistance.

Therefore it is suitable.

Borosilicate glass is suitable.

Common mistake: Always connect property to application.

Classify polymers

Given: Polythene, PVC, Nylon-66, Bakelite.

Polythene and PVC are addition polymers.

Nylon-66 and Bakelite are condensation polymers.

Polythene/PVC are thermoplastics; Bakelite is thermosetting.

Addition: polythene, PVC. Condensation: Nylon-66, Bakelite.

Common mistake: Learn examples as pairs.

Nanomaterial classification

Given: Nanoparticle, nanotube, graphene.

Nanoparticle has all dimensions nanoscale → 0D.

Nanotube has one extended dimension → 1D.

Graphene is a sheet → 2D.

Nanoparticle: 0D; nanotube: 1D; graphene: 2D.

Common mistake: Dimension classification is enough for this syllabus.

Expected Questions

Questions only are shown here. Full answers are available in Master Answer Key.

Q1 What is alloying? Explain purposes of alloying with examples brass, bronze and solder.

Q2 Explain types and applications of glasses in engineering.

Q3 What are refractory materials? Write characteristics and applications.

Q4 Define monomer, polymer and polymerization. Classify polymers.

Q5 Explain polythene, PVC, Nylon-66 and Bakelite with monomers and uses.

Q6 Compare thermoplastics and thermosetting plastics.

Q7 Explain natural rubber, vulcanization, Buna-S and Buna-N.

Q8 Define nanomaterials and classify 0D, 1D and 2D with examples and applications.

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Module 4 • CO4 • 12 hours

Electrochemistry and Corrosion

This module explains electron-transfer reactions, electrolysis, electroplating, refining, electrochemical cells and corrosion control methods.

Electron transfer reactions, electrolysis, electroplating, refining, electrochemical cells, corrosion prevention methods എന്നിവ ഈ module പഠിപ്പിക്കുന്നു.

M4.01 Oxidation, reduction, conductors and electrolytes

English explanation

Oxidation is the loss of electrons and reduction is the gain of electrons. A redox reaction is a reaction in which oxidation and reduction occur simultaneously. Metallic conductors conduct electricity due to movement of electrons, while electrolytic conductors conduct due to movement of ions. Electrolytes are substances that conduct electricity in molten or aqueous state and undergo chemical decomposition. Strong electrolytes ionize almost completely, while weak electrolytes ionize partially. Non-electrolytes do not conduct electricity in solution. Electrolysis is the chemical decomposition of an electrolyte by passing electric current. An electrolytic cell converts electrical energy into chemical change.

Malayalam support

Oxidation electron നഷ്ടപ്പെടൽ, reduction electron നേടൽ ആണ്. Redox reaction-ൽ oxidation ഉം reduction ഉം ഒരേസമയം നടക്കും. Metallic conductor-ൽ electrons move ചെയ്ത് current flow ചെയ്യും. Electrolytic conductor-ൽ ions move ചെയ്യും. Electrolyte molten/aqueous state-ൽ current conduct ചെയ്ത് chemical decomposition കാണിക്കും. Strong electrolytes almost complete ionization; weak electrolytes partial ionization. Non-electrolytes conduct ചെയ്യില്ല. Electric current ഉപയോഗിച്ച് electrolyte decompose ചെയ്യുന്നത് electrolysis ആണ്. Electrolytic cell electrical energy chemical change ആക്കുന്നു.

Redox memory

Oxidation
Loss of e^-

$\rightarrow$

Reduction
Gain of e^-

OIL RIG: Oxidation Is Loss, Reduction Is Gain

Important points

- Oxidation = loss of electrons.
- Reduction = gain of electrons.
- Strong electrolyte: HCl, NaOH, NaCl.
- Weak electrolyte: CH_3COOH, NH_4OH.
- Electrolysis needs electrolyte, electrodes and DC source.

Examples

- $Zn \rightarrow Zn^{2+} + 2e^-$ is oxidation.
- $Cu^{2+} + 2e^- \rightarrow Cu$ is reduction.

Applications

- Electroplating, refining, battery reactions and corrosion.

Study tricks / exam tips:

- Use OIL RIG memory: Oxidation Is Loss, Reduction Is Gain.
- In electrolytic cell, anode is positive and cathode is negative.

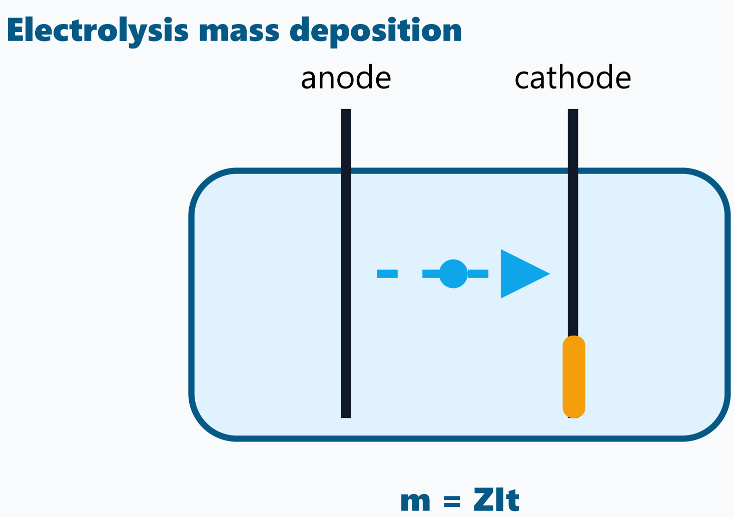
M4.02 Faraday's laws of electrolysis

English explanation

Faraday's first law states that the mass of substance deposited or liberated at an electrode is directly proportional to the quantity of electricity passed through the electrolyte. Since quantity of electricity $Q = It$, mass $m = ZIt$, where Z is electrochemical equivalent. Faraday's second law states that when the same quantity of electricity passes through different electrolytes, the masses deposited are proportional to their equivalent masses. In practical problems, time must be converted to seconds and current must be in ampere. Faraday constant, approximately 96500 C, is the charge required to deposit or liberate one gram equivalent of a substance.

Malayalam support

Faraday's first law പ്രകാരം electrode-ൽ deposited/liberated mass, electrolyte വഴി കടന്ന electric charge-നോട് directly proportional ആണ്. $Q = It$ ആയതിനാൽ $m = ZIt$. Z electrochemical equivalent. Faraday's second law പ്രകാരം same quantity electricity different electrolytes വഴി കടന്നാൽ deposited masses equivalent masses-നോട് proportional. Problems ചെയ്യുമ്പോൾ time seconds-ലും current ampere-ലും വേണം. Faraday constant ഏകദേശം 96500 C ആണ്; ഒരു gram equivalent deposit ചെയ്യാൻ വേണ്ട charge.



Important points

- First law: $m \propto Q$, $m = ZIt$.
- Second law: masses deposited are proportional to equivalent masses.
- $F = 96500$ C per equivalent.
- Equivalent mass = atomic mass / valency.

Examples

- Copper deposition during electrolysis of $CuSO_4$.
- Silver plating using controlled current and time.

Applications

- Electroplating thickness control.
- Electrolytic refining and metal deposition.

Study tricks / exam tips:

- Always convert minutes to seconds.
- Mention unit of current, time and mass in numerical answer.

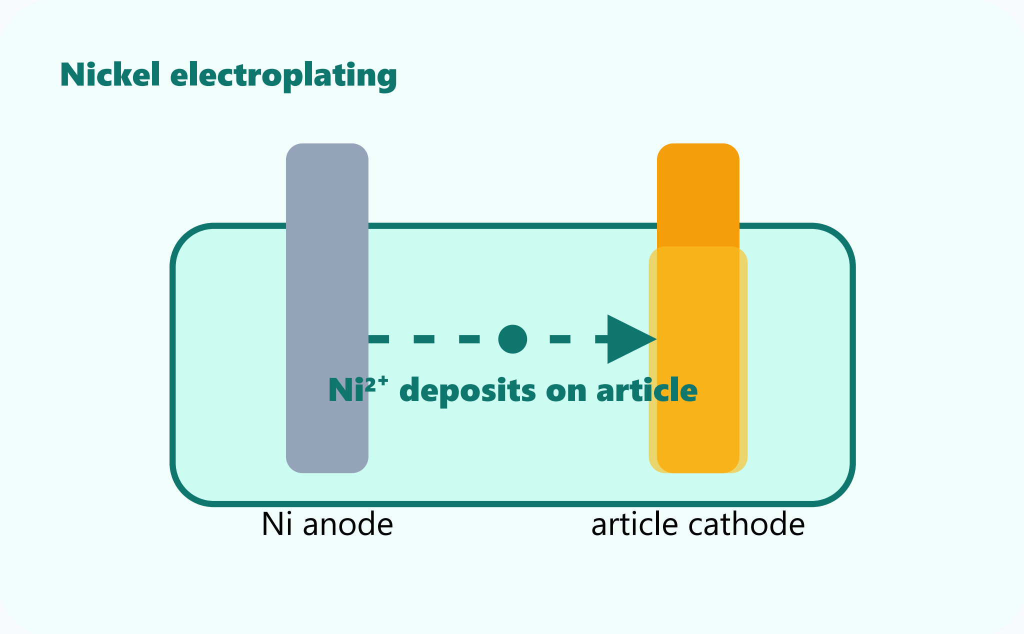
M4.03 Electroplating and electrolytic refining

English explanation

Electroplating is the process of depositing a thin layer of one metal over another metal by electrolysis. It improves appearance, corrosion resistance and surface hardness. In nickel plating on mild steel, the mild steel article is made cathode, pure nickel is made anode and nickel salt solution is used as electrolyte. When current passes, nickel dissolves from anode as Ni^{2+} and deposits on the cathode article. Electrolytic refining is used to purify metals. In copper refining, impure copper is the anode, pure copper sheet is cathode and acidified copper sulphate solution is electrolyte. Copper dissolves from anode and pure copper deposits on cathode, while impurities settle as anode mud.

Malayalam support

Electroplating electrolysis ഉപയോഗിച്ച് ഒരു metal-ന്റെ thin layer മറ്റൊരു metal-ന്റെ surface-ൽ deposit ചെയ്യുന്നതാണ്. Appearance, corrosion resistance, surface hardness എന്നിവ improve ചെയ്യും. Nickel plating on mild steel-ൽ mild steel article cathode, pure nickel anode, nickel salt solution electrolyte. Current pass ചെയ്യുമ്പോൾ nickel anode-ൽ നിന്ന് dissolve ചെയ്ത് cathode-ൽ deposit ചെയ്യും. Electrolytic refining metal purify ചെയ്യാൻ ഉപയോഗിക്കുന്നു. Copper refining-ൽ impure copper anode, pure copper cathode, acidified copper sulphate electrolyte. Pure copper cathode-ൽ deposit ചെയ്യുകയും impurities anode mud ആയി settle ചെയ്യുകയും ചെയ്യും.



Important points

- Electroplating article is cathode.
- Plating metal is usually anode.
- Electrolyte contains ions of plating metal.
- Refining: impure metal anode, pure metal cathode.

Examples

- Nickel plating on mild steel.
- Copper refining in industry.

Applications

- Corrosion protection, decorative coating, electrical contacts, metal purification.

Study tricks / exam tips:

- For electroplating diagram: label anode, cathode, electrolyte and DC source.
- Do not interchange article and anode in plating answer.

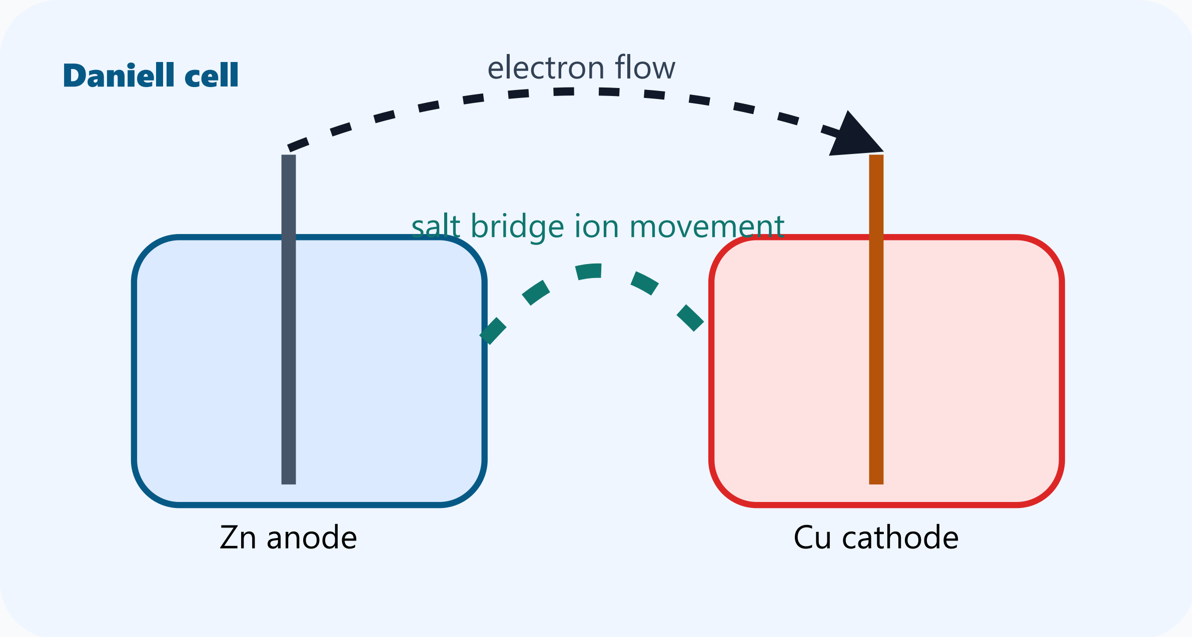
M4.04 Electrochemical cells and Daniell cell

English explanation

An electrochemical cell converts chemical energy into electrical energy through a redox reaction. Primary cells are not easily rechargeable, secondary cells are rechargeable, and fuel cells produce electricity as long as fuel and oxidant are supplied. Daniell cell is a classical example of an electrochemical cell. It consists of zinc electrode dipped in zinc sulphate solution and copper electrode dipped in copper sulphate solution, connected by a salt bridge. At zinc electrode, oxidation occurs: $Zn \rightarrow Zn^{2+} + 2e^-$. At copper electrode, reduction occurs: $Cu^{2+} + 2e^- \rightarrow Cu$. Electrons flow through the external circuit from zinc to copper. The salt bridge maintains electrical neutrality. Cell emf can be calculated using Ecell = Ecathode − Eanode.

Malayalam support

Electrochemical cell chemical energy electrical energy ആക്കി മാറ്റുന്നു. Primary cells recharge ചെയ്യാൻ സാധാരണയായി കഴിയില്ല; secondary cells rechargeable ആണ്; fuel cells fuel/oxidant supply ഉണ്ടായാൽ electricity produce ചെയ്യും. Daniell cell-ൽ zinc electrode zinc sulphate solution-ലും copper electrode copper sulphate solution-ലും ഇടുന്നു; salt bridge connect ചെയ്യും. Zinc electrode-ൽ oxidation: $Zn \rightarrow Zn^{2+} + 2e^-$. Copper electrode-ൽ reduction: $Cu^{2+} + 2e^- \rightarrow Cu$. Electrons external circuit വഴി zinc-ൽ നിന്ന് copper-ലേക്ക് flow ചെയ്യും. Salt bridge electrical neutrality maintain ചെയ്യുന്നു. Ecell = Ecathode − Eanode ഉപയോഗിച്ച് emf calculate ചെയ്യാം.



Important points

- Anode in Daniell cell: zinc; oxidation.
- Cathode: copper; reduction.
- Salt bridge completes circuit and maintains neutrality.
- Ecell = Ecathode − Eanode.

Examples

- $Zn|Zn^{2+}$ || $Cu^{2+}|Cu$ cell notation.
- Dry cell is primary; lead-acid battery is secondary; hydrogen-oxygen cell is fuel cell.

Applications

- Batteries, sensors and electrochemical measurements.

Study tricks / exam tips:

- In galvanic cell, anode is negative and cathode is positive.
- Write half-cell reactions separately before total reaction.

M4.05 Corrosion and corrosion prevention

English explanation

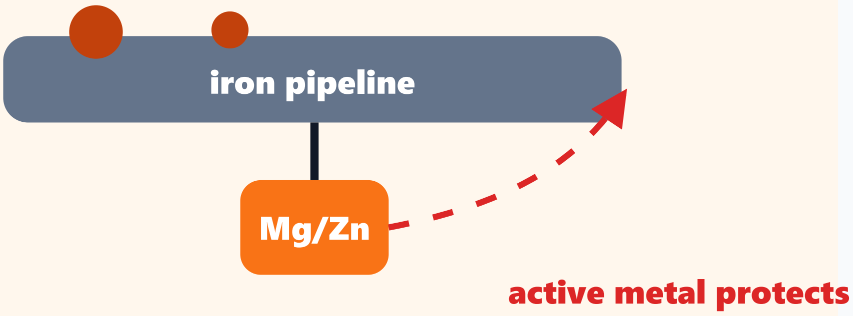
Corrosion is the gradual destruction of a metal by chemical or electrochemical reaction with the environment. Rusting of iron is a common example. Corrosion rate increases with moisture, oxygen, salts, pollutants, temperature and contact with dissimilar metals. It causes loss of strength, leakage, failure of machines, contamination and economic loss. Corrosion prevention can be done by barrier protection, metallic coatings, non-metallic coatings and cathodic protection. In barrier protection, paint, grease or polymer coating prevents contact between metal and environment. Metallic coatings may be anodic or cathodic depending on the coating metal. Anodising forms a protective oxide layer on aluminium. In sacrificial anode method, a more active metal such as zinc or magnesium is connected to the protected metal, so the active metal corrodes preferentially.

Malayalam support

Corrosion എന്നത് metal environment-നോട് chemical/electrochemical reaction ചെയ്ത് ധീരേ ധീരേ destroy ആകുന്നതാണ്. Iron rusting സാധാരണ example ആണ്. Moisture, oxygen, salts, pollutants, temperature, dissimilar metal contact എന്നിവ corrosion rate വർധിപ്പിക്കും. Strength കുറവ്, leakage, machine failure, contamination, economic loss എന്നിവ ഉണ്ടാകും. Prevention: barrier protection, metallic coating, non-metallic coating, cathodic protection. Paint/grease/polymer coating metal-നെ environment-ൽ നിന്ന് isolate ചെയ്യും. Anodising aluminium-ൽ protective oxide layer ഉണ്ടാക്കുന്നു. Sacrificial anode method-ൽ zinc/magnesium പോലുള്ള active metal protect ചെയ്യേണ്ട metal-നോട് connect ചെയ്യുന്നു; active metal ആദ്യം corrode ചെയ്യും.

Corrosion protection

Sacrificial anode corrodes first; pipeline becomes cathode.



Important points

- Corrosion is environmental destruction of metal.
- Factors: moisture, oxygen, salts, temperature, pollutants, dissimilar metals.
- Barrier protection prevents contact.
- Sacrificial anode protects by preferential corrosion.

Examples

- Rusting of iron bridge.
- Zinc coating on iron by galvanising.
- Magnesium anode protecting underground pipeline.

Applications

- Pipelines, ships, bridges, boilers, tanks and electrical equipment.

Study tricks / exam tips:

- Prevention answer should mention principle, not only names.
- For sacrificial anode, active metal corrodes; protected metal becomes cathode.

Solved Examples

Worked examples are included in the module itself. Master Answer Key contains answers for expected questions.

Faraday law mass deposition

Given: Current = 2 A, time = 30 min, Z = 0.000329 g/C.

$t = 30 \times 60 = 1800$ s.

$m = ZIt$

$m = 0.000329 \times 2 \times 1800 = 1.1844$ g

Mass deposited ≈ 1.18 g.

Common mistake: Time must be in seconds.

Daniell cell reactions

Given: Zn/Zn^{2+} and Cu^{2+}/Cu half cells.

At zinc anode: $Zn \rightarrow Zn^{2+} + 2e^-$.

At copper cathode: $Cu^{2+} + 2e^- \rightarrow Cu$.

Overall: $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$.

Zinc is anode, copper is cathode; electrons flow from Zn to Cu.

Common mistake: Oxidation at anode, reduction at cathode.

Corrosion prevention selection

Given: Underground iron pipeline must be protected.

Soil moisture causes electrochemical corrosion.

Connect a more active metal such as magnesium or zinc.

The active metal acts as sacrificial anode and corrodes instead.

Use cathodic protection by sacrificial anode method.

Common mistake: Protected metal behaves as cathode.

Expected Questions

Questions only are shown here. Full answers are available in Master Answer Key.

Q1 Explain electronic concept of oxidation, reduction and redox reaction with examples.

Q2 Differentiate metallic conductors, electrolytic conductors, electrolytes and non-electrolytes.

Q3 State Faraday’s first and second laws of electrolysis.

Q4 Solve a Faraday law problem for mass deposited.

Q5 Explain electroplating of nickel on mild steel.

Q6 Explain electrolytic refining of copper.

Q7 Explain Daniell cell with reactions and function of salt bridge.

Q8 Define corrosion, factors affecting corrosion and prevention methods.

Quick Revision Sheet

Last-minute checklist, study tricks and common mistakes for Applied Chemistry-I.

Study tricks

- Bond answer trick: Formation → electron transfer/sharing → example → property.
- Molarity trick: gram ÷ molar mass ÷ litre.
- pH trick: $[H^+] = 10^{-n}$ gives pH = n.
- Electrolysis trick: mass deposited increases when current or time increases.
- Corrosion answer trick: definition → factors → damage → prevention principle.

Common mistakes

- Using volume in mL directly in molarity formula.
- Writing pH high means more acidic; actually lower pH means more acidic.
- Confusing ionic and covalent bond examples.
- Forgetting time conversion from minutes to seconds in Faraday problems.
- Writing only names of water treatment steps without purpose.
- Interchanging anode and cathode in electroplating and Daniell cell.

Before exam

- Can I write all formula-bank equations?
- Can I draw Daniell cell and titration setup?
- Can I solve one pH, molarity and Faraday problem?
- Can I explain water treatment in order?
- Can I compare thermoplastic and thermosetting plastics?

Answer Writing Templates

Use these formats to make answers look like a textbook response.

5-mark theory answer

1. Definition
2. Principle or working
3. Labelled sketch/table where needed
4. Example
5. Application / advantage / limitation

Numerical answer

1. Given data with units
2. Formula
3. Substitution
4. Calculation
5. Final answer with unit

Material comparison answer

1. Composition or structure
2. Important properties
3. Two examples
4. Engineering applications
5. One limitation if asked

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Final Master Answer Key

These model answers match exactly with the Expected Questions inside each module.

Module 1 Answer Key

Atomic Structure and Chemical Bonding

Module 1 • Q1

Explain Bohr's atom model. Write its important postulates, merits and demerits.

Bohr proposed that electrons revolve around the nucleus only in certain permitted circular orbits called stationary states. In these orbits electrons do not radiate energy. Energy is absorbed or emitted only when an electron jumps between two energy levels. The model explains stability of atoms and hydrogen line spectrum. Its limitations are that it cannot explain spectra of multi-electron atoms, fine structure and Zeeman effect.

Module 1 • Q2

Define de Broglie wavelength and solve a simple problem using $\lambda = h/mv$.

de Broglie wavelength is the wavelength associated with a moving particle. It is given by $\lambda = h/mv$, where h is Planck constant, m is mass and v is velocity. In a problem, convert mass to kg and velocity to m/s, substitute in the formula and write final answer in metre. It shows wave nature of matter, especially microscopic particles.

Module 1 • Q3

State Heisenberg's uncertainty principle and explain its meaning.

Heisenberg's uncertainty principle states that it is impossible to determine simultaneously the exact position and exact momentum of a microscopic particle. Mathematically, $\Delta x \, \Delta p \geq h/4\pi$. It means the electron cannot be treated like a particle moving in a fixed path; instead probability and orbitals are used.

Module 1 • Q4

Explain quantum numbers and their significance.

Quantum numbers describe the state of an electron in an atom. Principal quantum number n indicates shell and energy. Azimuthal quantum number l indicates subshell and shape. Magnetic quantum number m indicates orientation of orbital. Spin quantum number s indicates spin direction. Together they help describe electron arrangement in orbitals.

Module 1 • Q5

Write Aufbau principle, Pauli exclusion principle and Hund's rule.

Aufbau principle: electrons fill lower energy orbitals before higher energy orbitals. Pauli exclusion principle: no two electrons in an atom can have the same set of four quantum numbers; an orbital can hold maximum two electrons with opposite spins. Hund's rule: degenerate orbitals are singly occupied before pairing occurs.

Module 1 • Q6

Write electronic configuration of first 20 elements using examples.

Electronic configuration is written by filling orbitals in order of increasing energy: 1s, 2s, 2p, 3s, 3p, 4s. Example: Na = $1s^2 2s^2 2p^6 3s^1$, Cl = $1s^2 2s^2 2p^6 3s^2 3p^5$, Ca = $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$. Shell configuration may also be written as 2,8,1 for Na and 2,8,8,2 for Ca.

Module 1 • Q7

Compare ionic, covalent and coordinate bonds with examples.

Ionic bond is formed by transfer of electrons, example NaCl. Covalent bond is formed by sharing of electrons, example H_2 and HF. Coordinate bond is formed when one atom donates both electrons of the shared pair, example NH_4^+. Ionic compounds usually have high melting point; covalent compounds often have lower melting points; coordinate bond behaves like covalent bond after formation.

Module 1 • Q8

Explain anomalous behaviour of water due to hydrogen bonding.

Water molecules form extensive hydrogen bonding because hydrogen is bonded to highly electronegative oxygen. This causes high boiling point, high surface tension, high specific heat and lower density of ice compared to liquid water. These properties are anomalous because H_2O behaves differently from similar hydrides.

Module 2 Answer Key

Analytical Chemistry and Water Treatment

Module 2 • Q1

Define solute, solvent and solution. Explain molarity, normality and ppm.

A solution is a homogeneous mixture of solute and solvent. Solute is the dissolved substance and solvent is the dissolving medium. Molarity is moles of solute per litre of solution. Normality is gram equivalents of solute per litre of solution. ppm means parts per million and for dilute water solutions it is approximately mg/L. These concentration units are used in titration and water analysis.

Module 2 • Q2

Solve a molarity problem from mass and volume.

Use the method: moles = given mass / molar mass, volume in litre = mL/1000, molarity = moles / volume in litre. Example: 4 g NaOH in 500 mL, molar mass 40 g/mol. Moles = 4/40 = 0.1 mol. Volume = 0.5 L. M = 0.1/0.5 = 0.2 M.

Module 2 • Q3

Explain pH, pOH and their relation. Add applications of pH.

pH = $-\log[H^+]$ and pOH = $-\log[OH^-]$. At 25°C, pH + pOH = 14. pH below 7 is acidic, pH 7 is neutral and pH above 7 is basic. pH control is important in water treatment, agriculture, medicine, electroplating, boiler water and corrosion prevention.

Module 2 • Q4

What is buffer solution? Give one acidic and one basic buffer example.

A buffer solution resists change in pH when small amounts of acid or base are added. Acidic buffer is made from a weak acid and its salt, for example acetic acid and sodium acetate. Basic buffer is made from a weak base and its salt, for example ammonium hydroxide and ammonium chloride.

Module 2 • Q5

Explain volumetric analysis, titration, end point and indicator.

Volumetric analysis is a quantitative method to find concentration using measured volumes of reacting solutions. Titration is the experimental process. End point is the stage where indicator changes colour showing completion of reaction. Indicator is a substance that shows colour change near the equivalence point. The choice of indicator depends on titration type.

Module 2 • Q6

Use normality equation to find unknown concentration.

At equivalence point, $N_1V_1 = N_2V_2$. Substitute known normality and volumes using the same volume unit. Example: 25 mL HCl is neutralized by 20 mL 0.1 N NaOH. N(HCl) × 25 = 0.1 × 20, so N(HCl) = 0.08 N.

Module 2 • Q7

Explain hardness of water and problems caused in boilers.

Hardness is due to calcium and magnesium salts in water. Temporary hardness is due to bicarbonates and permanent hardness is due to chlorides and sulphates. Hard water forms scum with soap and does not lather easily. In boilers it produces scale, reduces heat transfer, wastes fuel and causes corrosion.

Module 2 • Q8

Explain lime-soda process, ion exchange method and municipal water treatment steps.

Lime-soda process removes hardness by precipitating calcium and magnesium salts. Ion exchange method uses resins to replace Ca^{2+} and Mg^{2+} ions with Na^{+} or H^{+} ions. Municipal water treatment includes sedimentation, coagulation, filtration and sterilization by chlorination, bleaching powder or UV radiation to produce potable water.

Module 3 Answer Key

Engineering Materials, Polymers and Nanomaterials

Module 3 • Q1

What is alloying? Explain purposes of alloying with examples brass, bronze and solder.

Alloying is the process of mixing two or more elements, at least one metal, to improve properties. It improves strength, hardness, corrosion resistance, melting point and castability. Brass is copper-zinc alloy used in fittings. Bronze is copper-tin alloy used in bearings and statues. Solder is used for joining metals.

Module 3 • Q2

Explain types and applications of glasses in engineering.

Glass is an amorphous inorganic material. Sodium silicate glass is common glass used in windows and containers. Borosilicate glass resists thermal shock and is used in laboratory apparatus. Safety glass reduces injury during breakage and is used in vehicles. Insulating glass reduces heat transfer in buildings.

Module 3 • Q3

What are refractory materials? Write characteristics and applications.

Refractory materials withstand high temperatures without softening or reacting. Important characteristics are high melting point, thermal stability, chemical resistance, mechanical strength and low thermal expansion. Examples include fire clay bricks and silica bricks. They are used in furnaces, kilns, boilers and reactors.

Module 3 • Q4

Define monomer, polymer and polymerization. Classify polymers.

Monomer is a small molecule capable of joining with similar or different molecules. Polymer is a large molecule made of repeating monomer units. Polymerization is the process of forming polymer. Homopolymer is made from one monomer; copolymer from two or more monomers. Addition polymers form without by-product; condensation polymers form with elimination of small molecules.

Module 3 • Q5

Explain polythene, PVC, Nylon-66 and Bakelite with monomers and uses.

Polythene is formed from ethene and is used for films, containers and insulation. PVC is formed from vinyl chloride and used for pipes and cable insulation. Nylon-66 is formed from hexamethylenediamine and adipic acid and used for fibres and machine parts. Bakelite is formed from phenol and formaldehyde and used for switches and electrical fittings.

Module 3 • Q6

Compare thermoplastics and thermosetting plastics.

Thermoplastics soften on heating and harden on cooling, so they can be remoulded; examples are polythene and PVC. Thermosetting plastics harden permanently on heating and cannot be remoulded; examples are Bakelite and melamine. Thermosetting plastics are generally harder and more heat resistant.

Module 3 • Q7

Explain natural rubber, vulcanization, Buna-S and Buna-N.

Natural rubber is a polymer of isoprene and is elastic but weak and temperature sensitive. Vulcanization is heating rubber with sulphur to improve strength, elasticity, durability and resistance. Buna-S is made from butadiene and styrene and used in tyres. Buna-N is made from butadiene and acrylonitrile and is oil resistant.

Module 3 • Q8

Define nanomaterials and classify 0D, 1D and 2D with examples and applications.

Nanomaterials have at least one dimension in the range of about 1–100 nm. 0D nanomaterials include nanoparticles, 1D include nanotubes and nanowires, and 2D include graphene sheets. Carbon nanotubes may be single-walled or multi-walled. Applications include sensors, electronics, coatings, medicine, energy storage and composites.

Module 4 Answer Key

Electrochemistry and Corrosion

Module 4 • Q1

Explain electronic concept of oxidation, reduction and redox reaction with examples.

Oxidation is loss of electrons and reduction is gain of electrons. A redox reaction involves both processes simultaneously. Example: $Zn \rightarrow Zn^{2+} + 2e^-$ is oxidation and $Cu^{2+} + 2e^- \rightarrow Cu$ is reduction. In the overall reaction $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$, zinc is oxidized and copper ion is reduced.

Module 4 • Q2

Differentiate metallic conductors, electrolytic conductors, electrolytes and non-electrolytes.

Metallic conductors conduct by movement of electrons and do not undergo chemical change, example copper. Electrolytic conductors conduct by movement of ions and undergo chemical change, example NaCl solution. Electrolytes conduct in molten or aqueous state; strong electrolytes ionize almost completely and weak electrolytes partially. Non-electrolytes like sugar solution do not conduct.

Module 4 • Q3

State Faraday's first and second laws of electrolysis.

First law: mass deposited or liberated at an electrode is directly proportional to the quantity of electricity passed, $m = ZIt$. Second law: when the same quantity of electricity passes through different electrolytes, masses deposited are proportional to their equivalent masses. These laws are used in electroplating and refining calculations.

Module 4 • Q4

Solve a Faraday law problem for mass deposited.

Use $m = ZIt$ or $m = EIt/F$. Convert time to seconds and current to ampere. Example: $I = 2$ A, $t = 30$ min $= 1800$ s, $Z = 0.000329$ g/C. $m = 0.000329 \times 2 \times 1800 = 1.18$ g. Therefore mass deposited is about 1.18 g.

Module 4 • Q5

Explain electroplating of nickel on mild steel.

In nickel plating, the mild steel article is made cathode and pure nickel is made anode. Nickel salt solution is used as electrolyte. On passing current, nickel dissolves from anode as Ni^{2+} and deposits as a thin layer on the steel article at cathode. It improves appearance and corrosion resistance.

Module 4 • Q6

Explain electrolytic refining of copper.

In copper refining, impure copper is the anode, pure copper sheet is the cathode and acidified copper sulphate is the electrolyte. Copper dissolves from anode as Cu^{2+} and deposits as pure copper on cathode. Insoluble impurities collect below the anode as anode mud.

Module 4 • Q7

Explain Daniell cell with reactions and function of salt bridge.

Daniell cell has zinc electrode in zinc sulphate and copper electrode in copper sulphate connected by salt bridge. Zinc acts as anode and undergoes oxidation: $Zn \rightarrow Zn^{2+} + 2e^-$. Copper acts as cathode and undergoes reduction: $Cu^{2+} + 2e^- \rightarrow Cu$. Electrons flow from zinc to copper. Salt bridge maintains electrical neutrality and completes internal circuit.

Module 4 • Q8

Define corrosion, factors affecting corrosion and prevention methods.

Corrosion is the gradual destruction of metal by chemical or electrochemical reaction with environment. Moisture, oxygen, salts, pollutants, high temperature and contact with dissimilar metals increase corrosion. Prevention methods include barrier protection by paint or grease, metallic coatings, non-metallic coatings such as anodising, anti-rust solutions and cathodic protection by sacrificial anode method.

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